CS T51 – OPERATING SYSTEMS

UNIT – IV

Virtual Memory – Demand Paging – Process creation – Page Replacement – Allocation of frames – Thrashing - File Concept – Access Methods – Directory Structure – File System Mounting – File Sharing – Protection.

2 Marks

1. What is File Control Block? (APR '14)

The logical file system manages the directory structure to provide the file-organization module with the information the latter needs, given a symbolic file name. It maintains file structure via file control blocks.

A file control block **(FCB)** contains information about the file, including ownership, permissions, and location of the file contents. The logical file system is also responsible for protection and security.

2. Mention about external fragmentation [Apr 2014]

When total memory is enough available to a process but cannot be allocated because of memory blocks are very small. That means program size is more than any available memory hole. Generally, external fragmentation occurs in dynamic or variable size partitions. External fragmentation can be solved using compaction technique. Also, external fragmentation can be prevented by paging or segmentation mechanisms.

3. What are the advantages and disadvantages of access control? (NOV'14)

The access control approach has the advantage of enabling complex access methodologies. The main problem with access lists is their length. If we want to allow everyone to read a file, we must list all users with read access. This technique has two undesirable consequences:

- Constructing such a list may be a tedious and unrewarding task, especially if we do not know in advance the list of users in the system.
- The directory entry, previously of fixed size, now must be of variable size, resulting in more complicated space management.

4. State the functions of working set strategy model (NOV'14)

Working Set Model –This model is based on the concept of the Locality Model.
The basic principle states that if we allocate enough frames to a process to accommodate its current locality, it will only fault whenever it moves to some new locality. But if the allocated frames are lesser than the size of the current locality, the process is bound to thrash.

According to this model, based on a parameter A, the working set is defined as the set of pages in the most recent 'A' page references. Hence, all the actively used pages would always end up being a part of the working set.

5. What are the different accessing methods of a file? (APR'15) (Sep 2020) [Nov 2019]

The different types of accessing a file are:

- Sequential access: Information in the file is accessed sequentially
- Direct access: Information in the file can be accessed without any particular order.
- Other access methods: Creating index for the file, indexed sequential access method (ISAM) etc.

6. What is thrashing and what are the causes of thrashing ? (NOV '15) [May 2019][Nov 2017] [May 2018][Apr 2016][May 2019]

Thrashing

- Thrashing occurs when a Process is spending more time in paging or Swapping activities rather than its execution.
- In Thrashing, the state CPU is so much busy swapping that it cannot respond to the user program as much as it required.

The Causes of Thrashing

- Thrashing is caused by under allocation of the minimum number of pages required by a process, forcing it to continuously page fault. The system can detect thrashing by evaluating the level of CPU utilization as compared to the level of multiprogramming. It can be eliminated by reducing the level of multiprogramming. '
- Thrashing refers to an instance of high paging activity. This happens when it is spending more time paging instead of executing.

7. What is the principle of optimality? (NOV '15)

Optimality is the best solution that can be achieved. In page replacement algorithm, the best replacement technique can be achieved by using optimal page replacement algorithm. It is simply replaces the page that will not be used for the longest period of time.

8. What is demand paging? (APR '16, APR '17, NOV'18)

A demand paging mechanism is very much similar to a paging system with swapping where processes stored in the secondary memory and pages are loaded only on demand, not in advance.

So, when a context switch occurs, the OS never copy any of the old program's pages from the disk or any of the new program's pages into the main memory. Instead, it will start executing the new program after loading the first page and fetches the program's pages, which are referenced.

9. What are the major problems to implement demand paging? [Apr 2017]

The two major problems to implement demand paging is developing
a. Frame allocation algorithm
b. Page replacement algorithm.

10. Define Belady's anomaly problem[Apr 2017]

Bélády's anomaly is the name given to the phenomenon where increasing the number of page frames results in an increase in the number of page faults for a given memory access pattern.

This phenomenon is commonly experienced in following page replacement algorithms:

- First in first out (FIFO)
- Second chance algorithm
- Random page replacement algorithm

Reason of Belady's Anomaly –

The other two commonly used page replacement algorithms are Optimal and LRU, but Belady'sAnamoly can never occur in these algorithms for any reference string as they belong to a class of stack based page replacement algorithms.

11. How is the directory structure used in File systems [Nov 2017]

- A File system contains thousands and millions of files, owned by several users. The directory structure organizes these files by keeping entries of all the related files. The file entries have information like file name, type, location, the mode in which the file can be accessed by other users in the system.

- A tree structure is the most common directory structure. The tree has a root directory, and every file in the system has a unique path.

Advantages:

- Very general, since full pathname can be given.
- Very scalable, the probability of name collision is less.
- Searching becomes very easy, we can use both absolute paths as well as relative.

12. Define Virtual Memory [May 2018]

Virtual memory is a technique that allows the execution of processes that may not be completely in memory. It is the separation of user logical memory from physical memory. This separation provides an extremely large virtual memory, when only a smaller physical memory is available.

13. what is mounting? (NOV '18)

Mounting is a process by which the operating system makes files and directories on a storage device (such as hard drive, CD-ROM, or network share) available for users to access via the computer's file system

.Mounting a file system attaches that file system to a directory (mount point) and makes it available to the system. The root (/) file system is always mounted. Any other file system can be connected or disconnected from the root (/) file system.

14. Under what circumstances do page faults occur [May 2019]

A page fault occurs when a program attempts to access a block of memory that is not stored in the physical memory, or RAM. The fault notifies the operating system that it must locate the data in virtual memory, then transfer it from the storage device, such as an HDD or SSD, to the system RAM.

15. What do you mean by best fit? [Nov 2019]

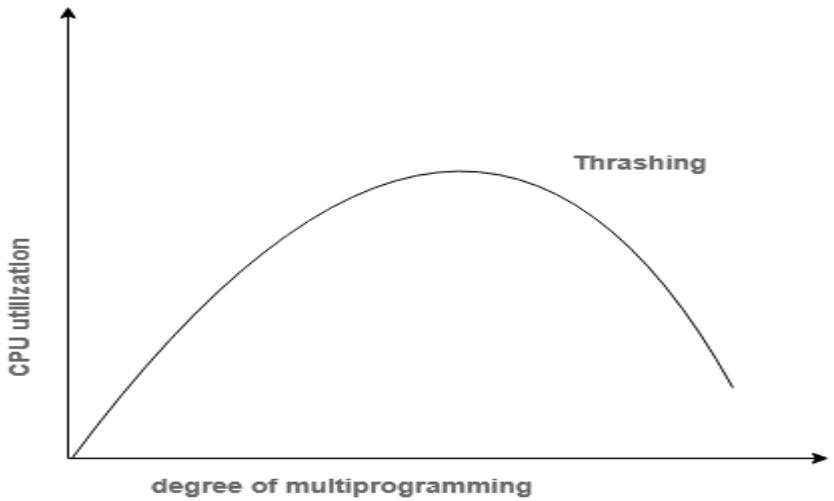
The **best fit** deals with allocating the smallest free partition which meets the requirement of the requesting process. It then tries to find a hole which is close to actual process size needed.

Assigning memory blocks to the process is challenging as the primary memory is needed to be divided among the **operating system**, user process, and the **operating system** process. Therefore, the system uses different algorithms to allocate memory from the main memory segment.

11 MARKS
Part 1

1. **Explain Thrashing? What are the causes of Thrashing? Explain about various impacts on CPU performance by Thrashing (APR '14) (NOV '15) (Nov 2016) [Nov 2019]**

   Thrashing is a condition or a situation when the system is spending a major portion of its time in servicing the page faults, but the actual processing done is very negligible.



In case, if the page fault and swapping happen very frequently at a higher rate, then the operating system has to spend more time swapping these pages. This state in the operating system is termed thrashing. Because of thrashing the CPU utilization is going to be reduced.

Causes of Thrashing

Thrashing affects the performance of execution in the Operating system. Also, thrashing results in severe performance problems in the Operating system.

When the utilization of CPU is low, then the process scheduling mechanism tries to load many processes into the memory at the same time due to which degree of Multiprogramming can be increased. Now in this situation, there are more processes in the memory as compared to the available number of frames in the memory. Allocation of the limited amount of frames to each process.

Whenever any process with high priority arrives in the memory and if the frame is not freely available at that time then the other process that has occupied the frame is residing in the frame will move to secondary storage and after that this free frame will be allocated to higher priority process.

We can also say that as soon as the memory fills up, the process starts spending a lot of time for the required pages to be swapped in. Again the utilization of the CPU becomes low because most of the processes are waiting for pages.

Thus a high degree of multiprogramming and lack of frames are two main causes of thrashing in the Operating system.

Effect of Thrashing

At the time, when thrashing starts then the operating system tries to apply either the **Global page replacement** Algorithm or the **Local page replacement** algorithm.
Global Page Replacement

The Global Page replacement has access to bring any page, whenever thrashing found it tries to bring more pages. Actually, due to this, no process can get enough frames and as a result, the thrashing will increase more and more. Thus, the global page replacement algorithm is not suitable whenever thrashing happens.

Local Page Replacement

Unlike the Global Page replacement, the local page replacement will select pages which only belongs to that process. Due to this, there is a chance of a reduction in the thrashing. As it is also proved that there are many disadvantages of Local Page replacement. Thus, local page replacement is simply an alternative to Global Page replacement.

Techniques used to handle the thrashing

Local Page replacement is better than the Global Page replacement but local page replacement has many disadvantages too, so it is not suggestible. Thus, given below are some other techniques that are used:

Working-Set Model

This model is based on the assumption of the locality. It makes the use of the parameter in order to define the working-set window. The main idea is to examine the most recent page reference. What locality is saying, the recently used page can be used again, and also the pages that are nearby this page will also be used

1. Working Set

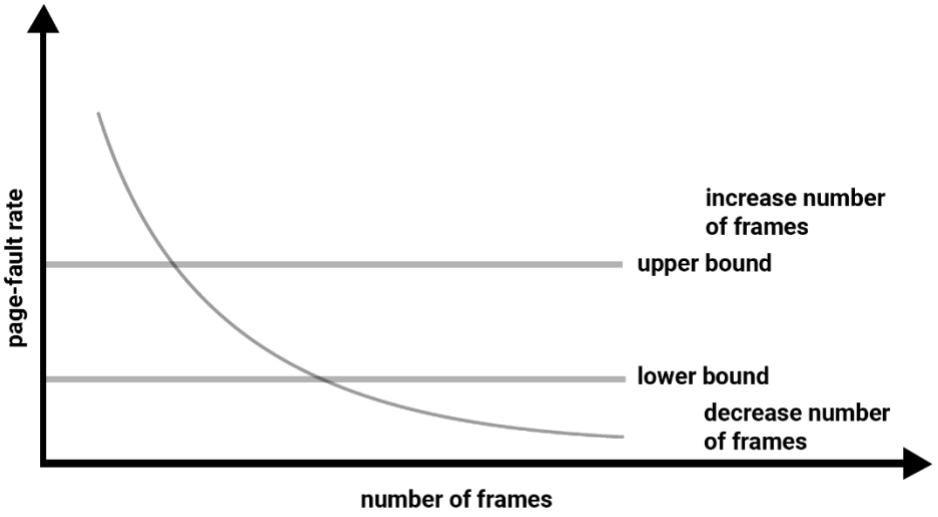
The set of the pages in the most recent? page reference is known as the working set. If a page is in active use, then it will be in the working set. In case if the page is no longer being used then it will drop from the working set ? times after its last reference.

The working set mainly gives the approximation of the locality of the program.

The accuracy of the working set mainly depends on what is chosen

This working set model avoids thrashing while keeping the degree of multiprogramming as high as possible.

2. Page Fault Frequency



The working-set model is successful and its knowledge can be useful in preparing but it is a very clumpy approach in order to avoid thrashing. There is another technique that is used to avoid thrashing and it is Page Fault Frequency(PFF) and it is a more direct approach.

The main problem is how to prevent thrashing. As thrashing has a high page fault rate and also we want to control the page fault rate.

When the Page fault is too high, then we know that the process needs more frames. Conversely, if the page fault-rate is too low then the process may have too many frames.

We can establish upper and lower bounds on the desired page faults. If the actual page-fault rate exceeds the upper limit then we will allocate the process to another frame. And if the page fault rate falls below the lower limit then we can remove the frame from the process.

Thus with this, we can directly measure and control the page fault rate in order to prevent thrashing.

2. Explain page replacement in detail? (APR'15) (NOV '15, NOV '18)

What is page replacement? Write down the basic replacement algorithm. How many page faults occur for FIFO algorithm for the following reference string: with four page frame (Reference String : 1,2,3,4,5,3,4,1,6,7,8,7,8,1,2) [Nov 2016] [Nov 2017] [Sep 2020]

Develop page faults and success given that main memory composed of three page frames for public use and that a program request pages in the follow order A,B,A,C,D,A,B,D,B,A,C,A,C,D using FIFO and LRU page removal algorithms. Do a page trace analysis [7] [Apr 2017]
Explain FIFO and LRU page replacement algorithm with example [Nov 2018] [Apr 2016]

Elucidate FIFO , Optimal and LRU page replacement techniques for the following 2,3,2,1,5,2,4,5,3,2,5,2

A page-replacement algorithm should minimize the number of page faults. We can do this minimization by distributing heavily used pages evenly over all of memory, rather than having them compete for a small number of page frames. We can associate with each page frame a counter of the number of pages that are associated with that frame. Then, to replace a page, we search for the page frame with the smallest counter. [May 2019]

Define a page-replacement algorithm using this basic idea. Specifically address the problems of (1) what the initial value of the counters is, (2) when counters are increased, (3) when counters are decreased, and (4) how the page to be replaced is selected.
a. How many page faults occur for your algorithm for the following reference string, for four page frames? 1, 2, 3, 4, 5, 3, 4, 1, 6, 7, 8, 7, 8, 9, 7, 8, 9, 5, 4, 5, 4, 2.
b. What is the minimum number of page faults for an optimal page replacement strategy for the reference string in part b with four page frames?

In an operating system that uses paging for memory management, a page replacement algorithm is needed to decide which page needs to be replaced when new page comes in.

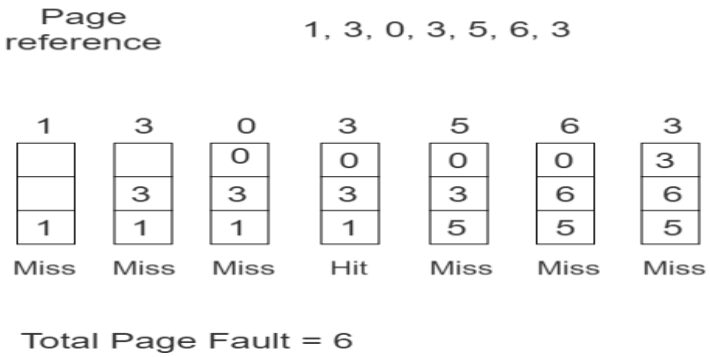
Page Fault – A page fault happens when a running program accesses a memory page that is mapped into the virtual address space, but not loaded in physical memory.
Since actual physical memory is much smaller than virtual memory, page faults happen. In case of page fault, Operating System might have to replace one of the existing pages with the newly needed page. Different page replacement algorithms suggest different ways to decide which page to replace. The target for all algorithms is to reduce the number of page faults.

Page Replacement Algorithms :

First In First Out (FIFO) –

This is the simplest page replacement algorithm. In this algorithm, the operating system keeps track of all pages in the memory in a queue, the oldest page is in the front of the queue. When a page needs to be replaced page in the front of the queue is selected for removal.

Example-1 Consider page reference string 1, 3, 0, 3, 5, 6 with 3 page frames.Find number of page faults.



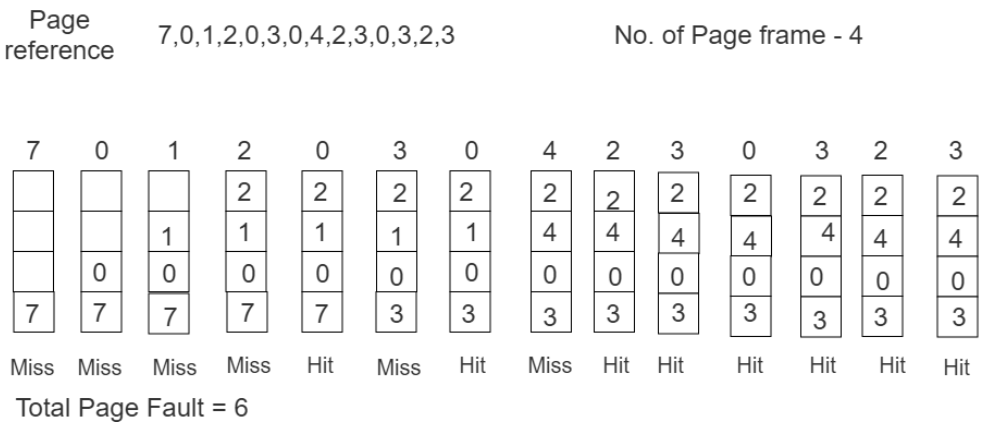
Initially all slots are empty, so when 1, 3, 0 came they are allocated to the empty slots —> **3 Page Faults.**
when 3 comes, it is already in memory so —> **0 Page Faults.**
Then 5 comes, it is not available in memory so it replaces the oldest page slot i.e 1. —>**1 Page Fault.**
6 comes, it is also not available in memory so it replaces the oldest page slot i.e 3 —>**1 Page Fault.**
Finally when 3 come it is not avilable so it replaces 0 **1 page fault**

Belady's anomaly – Belady's anomaly proves that it is possible to have more page faults when increasing the number of page frames while using the First in First Out (FIFO) page replacement algorithm. For example, if we consider reference string 3, 2, 1, 0, 3, 2, 4, 3, 2, 1, 0, 4 and 3 slots, we get 9 total page faults, but if we increase slots to 4, we get 10 page faults.

Optimal Page replacement –

In this algorithm, pages are replaced which would not be used for the longest duration of time in the future.

Example-2: Consider the page references 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, with 4 page frame. Find number of page fault.

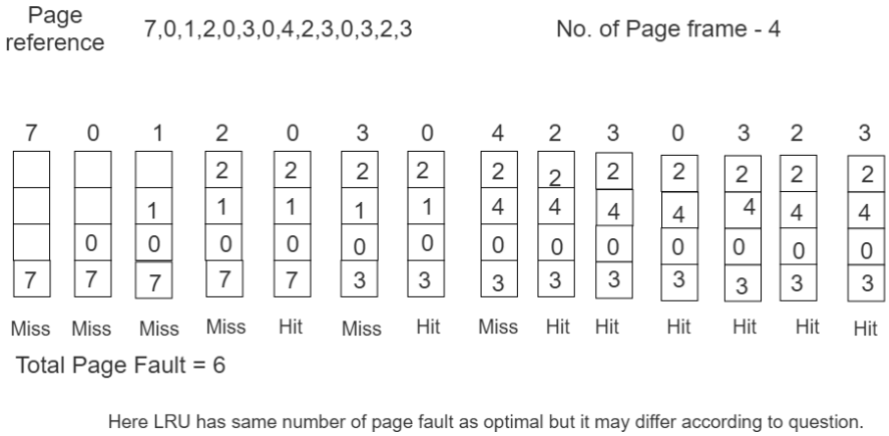


Initially all slots are empty, so when 7 0 1 2 are allocated to the empty slots —> **4 Page faults**
0 is already there so —> **0 Page fault.**
when 3 came it will take the place of 7 because it is not used for the longest duration of time in the future.—>**1 Page fault.**
0 is already there so —> **0 Page fault.**
4 will takes place of 1 —> **1 Page Fault.**

Now for the further page reference string —> **0 Page fault** because they are already available in the memory.

Optimal page replacement is perfect, but not possible in practice as the operating system cannot know future requests. The use of Optimal Page replacement is to set up a benchmark so that other replacement algorithms can be analyzed against it.

Least Recently Used –
In this algorithm page will be replaced which is least recently used.
Example-3 Consider the page reference string 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2 with 4 page frames.Find number of page faults.



Here LRU has same number of page fault as optimal but it may differ according to question.

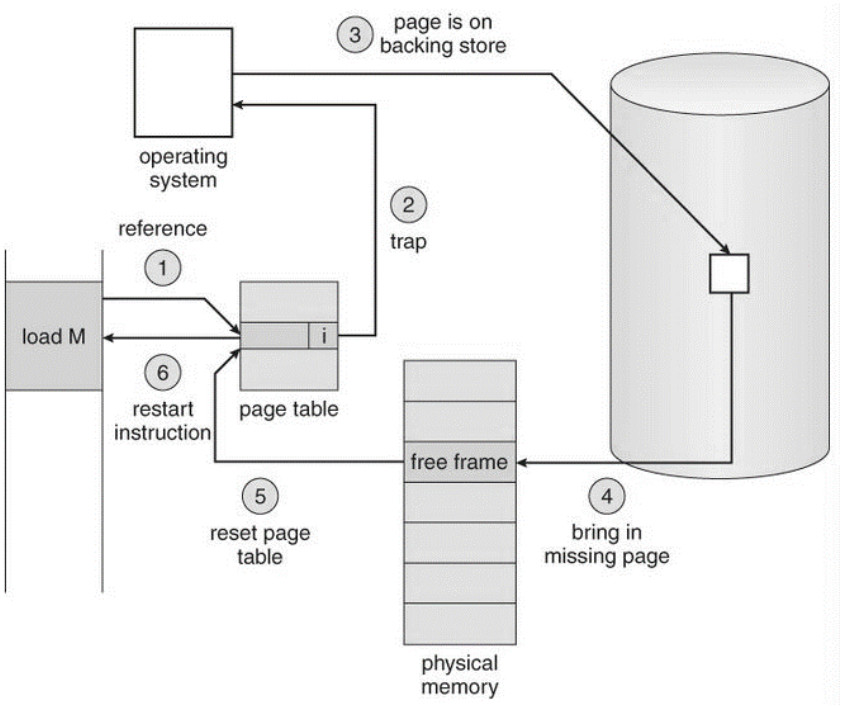
Initially all slots are empty, so when 7 0 1 2 are allocated to the empty slots —> **4 Page faults**

0 is already their so —> **0 Page fault.**
when 3 came it will take the place of 7 because it is least recently used —>**1 Page fault**
0 is already in memory so —> **0 Page fault**.
4 will takes place of 1 —> **1 Page Fault**

Now for the further page reference string —> **0 Page fault** because they are already available in the memory.

3. Write the procedure for handling page faults [4] [Apr 2017]

A page fault occurs when a program attempts to access data or code that is in its address space, but is not currently located in the system RAM. So when page fault occurs then following sequence of events happens :



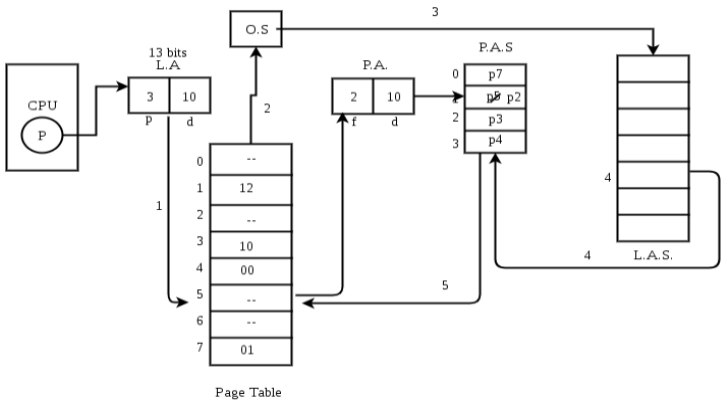
- The computer hardware traps to the kernel and program counter (PC) is saved on the stack. Current instruction state information is saved in CPU registers.
- An assembly program is started to save the general registers and other volatile information to keep the OS from destroying it.
- Operating system finds that a page fault has occurred and tries to find out which virtual page is needed. Sometimes hardware register contains this required information. If not, the operating system must retrieve PC, fetch instruction and find out what it was doing when the fault occurred.

- Once virtual address caused page fault is known, system checks to see if address is valid and checks if there is no protection access problem.
- If the virtual address is valid, the system checks to see if a page frame is free. If no frames are free, the page replacement algorithm is run to remove a page.
- If frame selected is dirty, page is scheduled for transfer to disk, context switch takes place, fault process is suspended and another process is made to run until disk transfer is completed.
- As soon as page frame is clean, operating system looks up disk address where needed page is, schedules disk operation to bring it in.
- When disk interrupt indicates page has arrived, page tables are updated to reflect its position, and frame marked as being in normal state.
- Faulting instruction is backed up to state it had when it began and PC is reset. Faulting is scheduled, operating system returns to routine that called it.
- Assembly Routine reloads register and other state information, returns to user space to continue execution.

Explain Demand paging [Nov 2019]

Demand Paging

The process of loading the page into memory on demand (whenever page fault occurs) is known as demand paging.The process includes the following steps :



1. If CPU try to refer a page that is currently not available in the main memory, it generates an interrupt indicating memory access fault.
2. The OS puts the interrupted process in a blocking state. For the execution to proceed the OS must bring the required page into the memory.
3. The OS will search for the required page in the logical address space.
4. The required page will be brought from logical address space to physical address space. The page replacement algorithms are used for the decision making of replacing the page in physical address space.
5. The page table will updated accordingly.

6. The signal will be sent to the CPU to continue the program execution and it will place the process back into ready state.
7. Hence whenever a page fault occurs these steps are followed by the operating system and the required page is brought into memory.

Advantages :

- More processes may be maintained in the main memory: Because we are going to load only some of the pages of any particular process, there is room for more processes. This leads to more efficient utilization of the processor because it is more likely that at least one of the more numerous processes will be in the ready state at any particular time.
- A process may be larger than all of main memory: One of the most fundamental restrictions in programming is lifted. A process larger than the main memory can be executed because of demand paging. The OS itself loads pages of a process in main memory as required.
- It allows greater multiprogramming levels by using less of the available (primary) memory for each process.

4. What are files and explain the access methods for files? (APR '14) (NOV'15) [Apr 2017][May 2019]

When a file is used, information is read and accessed into computer memory and there are several ways to access this information of the file. Some systems provide only one access method for files. Other systems, such as those of IBM, support many access methods, and choosing the right one for a particular application is a major design problem.

There are three ways to access a file into a computer system: Sequential-Access, Direct Access, Index sequential Method.

1. **Sequential Access –**

It is the simplest access method. Information in the file is processed in order, one record after the other. This mode of access is by far the most common; for example, editor and compiler usually access the file in this fashion.
Read and write make up the bulk of the operation on a file. A read operation *-read next-* read the next position of the file and automatically advance a file pointer, which keeps track I/O location. Similarly, for the write*write next* append to the end of the file and advance to the newly written material.
Key points:

- Data is accessed one record right after another record in an order.
- When we use read command, it move ahead pointer by one
- When we use write command, it will allocate memory and move the pointer to the end of the file

Such a method is reasonable for tape.

DirectAccess–

Another method is direct access method also known as relative access method. A filed-length logical record that allows the program to read and write record rapidly.

in no particular order. The direct access is based on the disk model of a file since disk allows random access to any file block. For direct access, the file is viewed as a numbered sequence of block or record. Thus, we may read block 14 then block 59 and then we can write block 17. There is no restriction on the order of reading and writing for a direct access file. A block number provided by the user to the operating system is normally a relative block number, the first relative block of the file is 0 and then 1 and so on.

Indexsequentialmethod–

It is the other method of accessing a file which is built on the top of the sequential access method. These methods construct an index for the file. The index, like an index in the back of a book, contains the pointer to the various blocks. To find a record in the file, we first search the index and then by the help of pointer we access the file directly.

Key points:

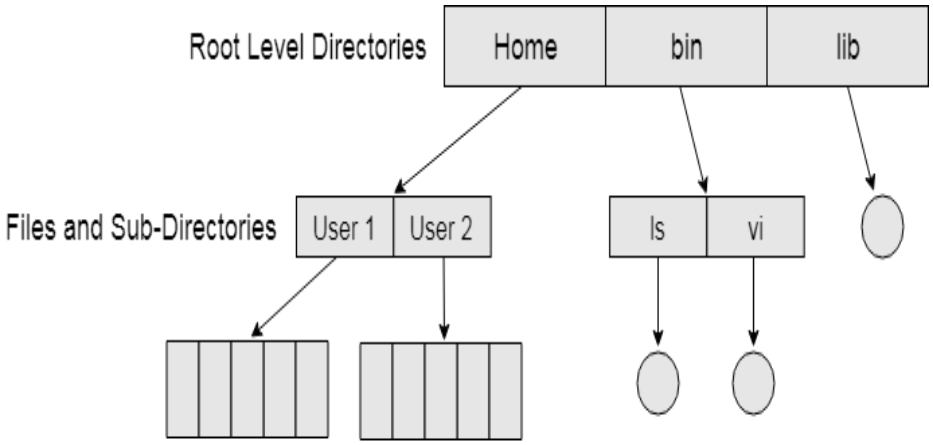
- It is built on top of Sequential access.
- It control the pointer by using index.

5. Explain Tree structured directories? (APR '14)

Tree Structured Directory

In Tree structured directory system, any directory entry can either be a file or sub directory. Tree structured directory system overcomes the drawbacks of two level directory system. The similar kind of files can now be grouped in one directory. Each user has its own directory and it cannot enter in the other user's directory. However, the user has the permission to read the root's data but he cannot write or modify this. Only administrator of the system has the complete access of root directory. Searching is more efficient in this directory structure. The concept of current working directory is used. A file can be accessed by two types of path, either relative or absolute.

Absolute path is the path of the file with respect to the root directory of the system while relative path is the path with respect to the current working directory of the system. In tree structured directory systems, the user is given the privilege to create the files as well as directories.



The Structured Directory System

Permissions on the file and directory

A tree structured directory system may consist of various levels therefore there is a set of permissions assigned to each file and directory.The permissions are **R W** X which are regarding reading, writing and the execution of the files or directory. The permissions are assigned to three types of users: owner, group and others. There is a identification bit which differentiate between directory and file. For a directory, it is d and for a file, it is dot **(.)**

6. Write notes about the protection strategies provided for files. (APR'15) [Sep 2020] [Nov 2017]

In computer systems, alot of user's information is stored, the objective of the operating system is to keep safe the data of the user from the improper access to the system. Protection can be provided in number of ways. For a single laptop system, we might provide protection by locking the computer in a desk drawer or file cabinet. For multi-user systems, different mechanisms are used for the protection.

- **Types of Access :**

  The files which have direct access of the any user have the need of protection.

- The files which are not accessible to other users doesn't require any kind of protection.
- The mechanism of the protection provide the facility of the controlled access by just limiting the types of access to the file.

Access can be given or not given to any user depends on several factors, one of which is the type of access required. Several different types of operations can be controlled:

- **Read –**Reading from a file.
- **Write –**Writing or rewriting the file.

- **Execute –**Loading the file and after loading the execution process starts.
- **Append –**Writing the new information to the already existing file, editing must be end at the end of the existing file.
- **Delete –**Deleting the file which is of no use and using its space for another data.
- **List –**List the name and attributes of the file.

Operations like renaming, editing the existing file, copying; these can also be controlled. There are many protection mechanism. each of them mechanism have different advantages and disadvantages and must be appropriate for the intended application.

Access Control :

There are different methods used by different users to access any file. The general way of protection is to associate *identity-dependent access* with all the files and directories an list called access-control list (ACL) which specify the names of the users and the types of access associate with each of the user. The main problem with the access list is their length. If we want to allow everyone to read a file, we must list all the users with the read access. This technique has two undesirable consequences:

- Constructing such a list may be tedious and unrewarding task, especially if we do not know in advance the list of the users in the system. Previously, the entry of the any directory is of the fixed size but now it changes to the variable size which results in the complicates space management.
- These problems can be resolved by use of a condensed version of the access list.

To condense the length of the access-control list, many systems recognize three classification of users in connection with each file:

- **Owner –**Owner is the user who has created the file.
- **Group –**A group is a set of members who has similar needs and they are sharing the same file.
- **Universe –**In the system, all other users are under the category called universe.

The most common recent approach is to combine access-control lists with the normal general owner, group, and universe access control scheme. For example: Solaris uses the three categories of access by default but allows access-control lists to be added to specific files and directories when more fine-grained access control is desired.

Other Protection Approaches:

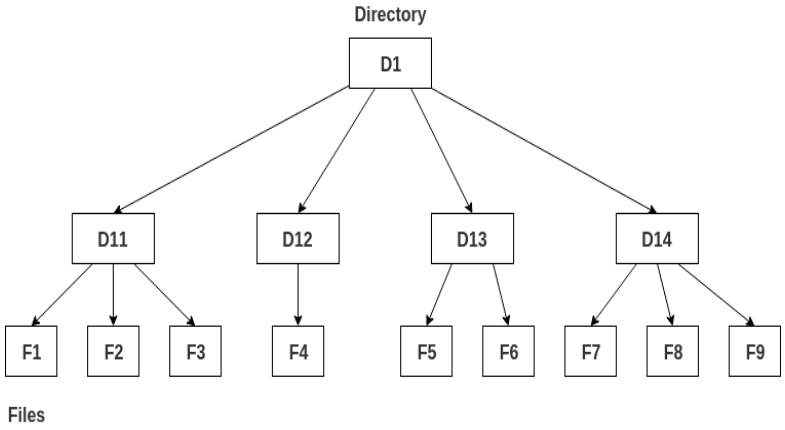
The access to any system is also controlled by the password. If the use of password are is random and it is changed often, this may be result in limit the effective access to a file.

The use of passwords has a few disadvantages:

- The number of passwords are very large so it is difficult to remember the large passwords.
- If one password is used for all the files, then once it is discovered, all files are accessible; protection is on all-or-none basis.

7. Explain about various directory structures [Apr 2016]

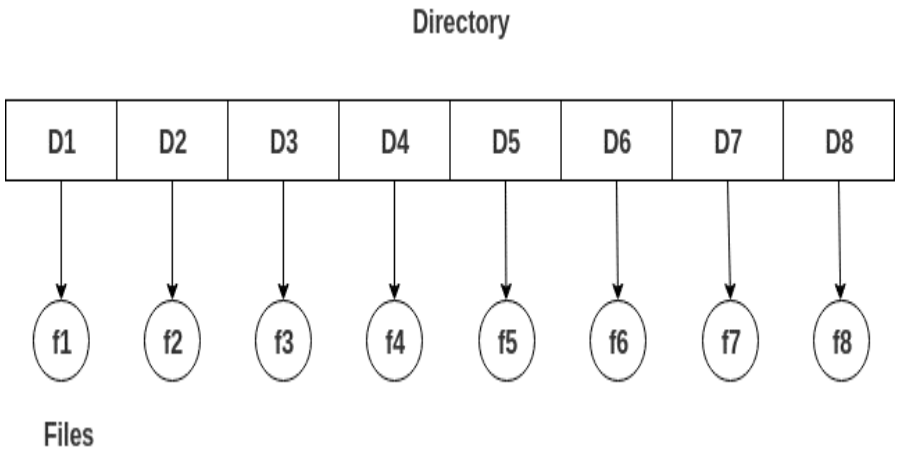
A **directory** is a container that is used to contain folders and files. It organizes files and folders in a hierarchical manner.



There are several logical structures of a directory, these are given below.

Single-leveldirectory–

The single-level directory is the simplest directory structure. In it, all files are contained in the same directory which makes it easy to support and understand. A single level directory has a significant limitation, however, when the number of files increases or when the system has more than one user. Since all the files are in the same directory, they must have a unique name. if two users call their dataset test, then the unique name rule violated.



Advantages:

- Since it is a single directory, so its implementation is very easy.

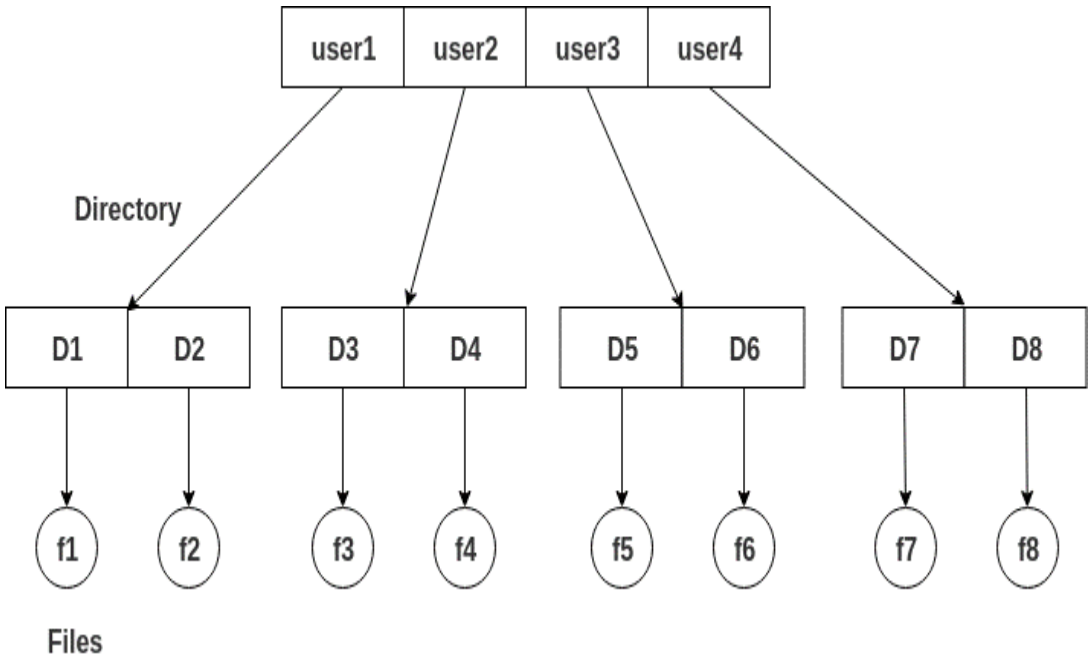
- If the files are smaller in size, searching will become faster.
- The operations like file creation, searching, deletion, updating are very easy in such a directory structure.

Disadvantages:

- There may chance of name collision because two files can not have the same name.
- Searching will become time taking if the directory is large.
- This can not group the same type of files together.

Two-leveldirectory–

As we have seen, a single level directory often leads to confusion of files names among different users. the solution to this problem is to create a separate directory for each user. In the two-level directory structure, each user has their own user files directory (UFD). The UFDs have similar structures, but each lists only the files of a single user. system's master file directory (MFD) is searches whenever a new user id=s logged in. The MFD is indexed by username or account number, and each entry points to the UFD for that user.



Advantages:

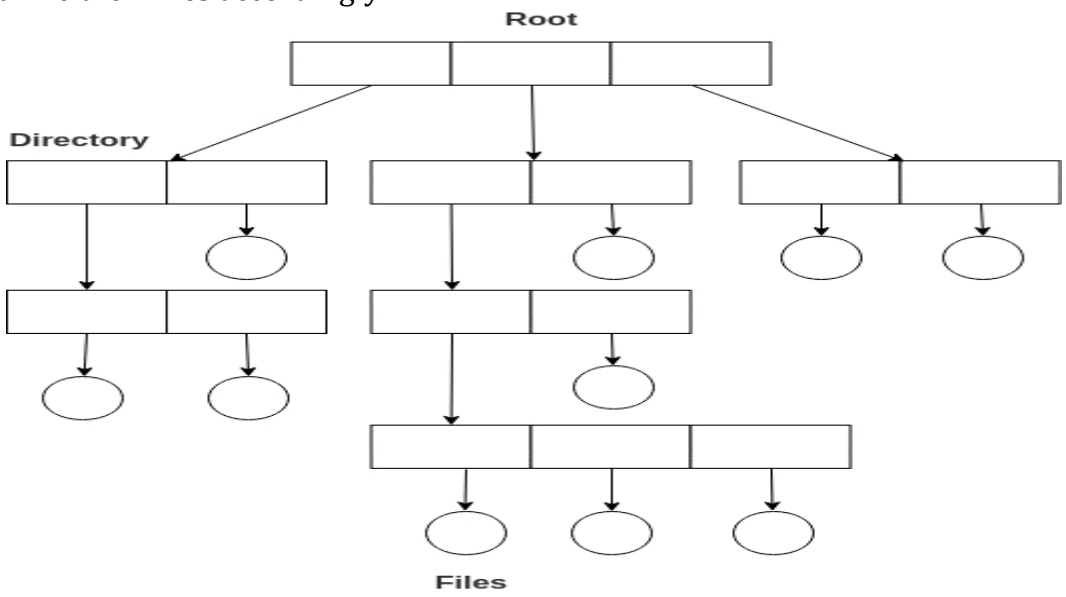
- We can give full path like /User-name/directory-name/.
- Different users can have the same directory as well as the file name.
- Searching of files becomes easier due to pathname and user-grouping.

Disadvantages:

- A user is not allowed to share files with other users.
- Still, it not very scalable, two files of the same type cannot be grouped together in the same user.

Tree-structured directory –

Once we have seen a two-level directory as a tree of height 2, the natural generalization is to extend the directory structure to a tree of arbitrary height. This generalization allows the user to create their own subdirectories and to organize their files accordingly.



A tree structure is the most common directory structure. The tree has a root directory, and every file in the system has a unique path.

Advantages:

- Very general, since full pathname can be given.
- Very scalable, the probability of name collision is less.
- Searching becomes very easy, we can use both absolute paths as well as relative.

Disadvantages:

- Every file does not fit into the hierarchical model, files may be saved into multiple directories.
- We can not share files.

It is inefficient, because accessing a file may go under multiple directories.

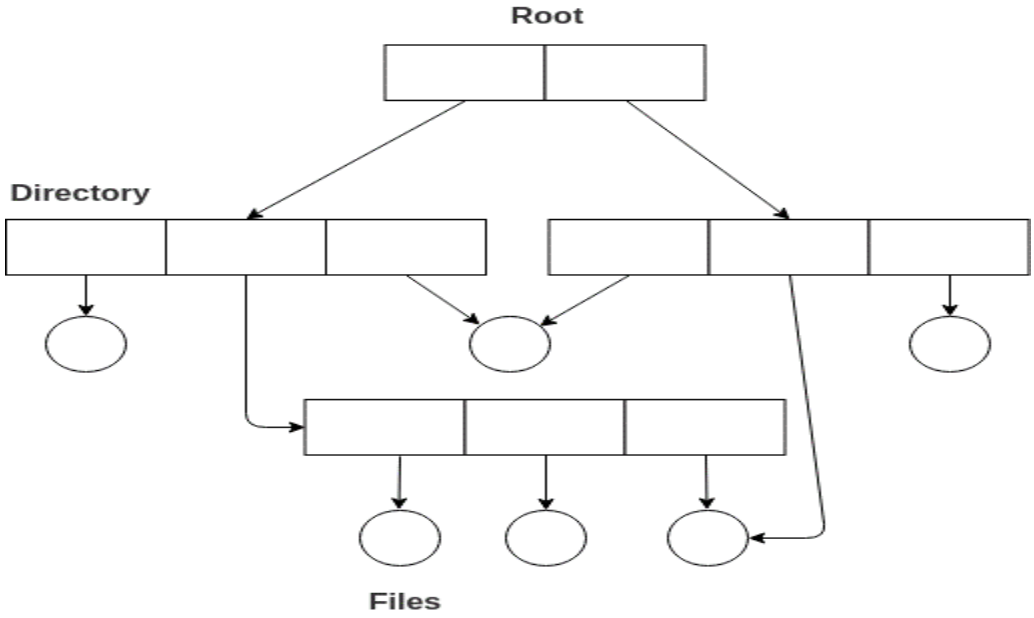
Acyclicgraphdirectory–

An acyclic graph is a graph with no cycle and allows us to share subdirectories and files. The same file or subdirectories may be in two different directories. It is a natural generalization of the tree-structured directory.

It is used in the situation like when two programmers are working on a joint project and they need to access files. The associated files are stored in a subdirectory, separating them from other projects and files of other programmers since they are working on a joint project so they want the subdirectories to be into their own

directories. The common subdirectories should be shared. So here we use Acyclic directories.

It is the point to note that the shared file is not the same as the copy file. If any programmer makes some changes in the subdirectory it will reflect in both subdirectories.



Advantages:

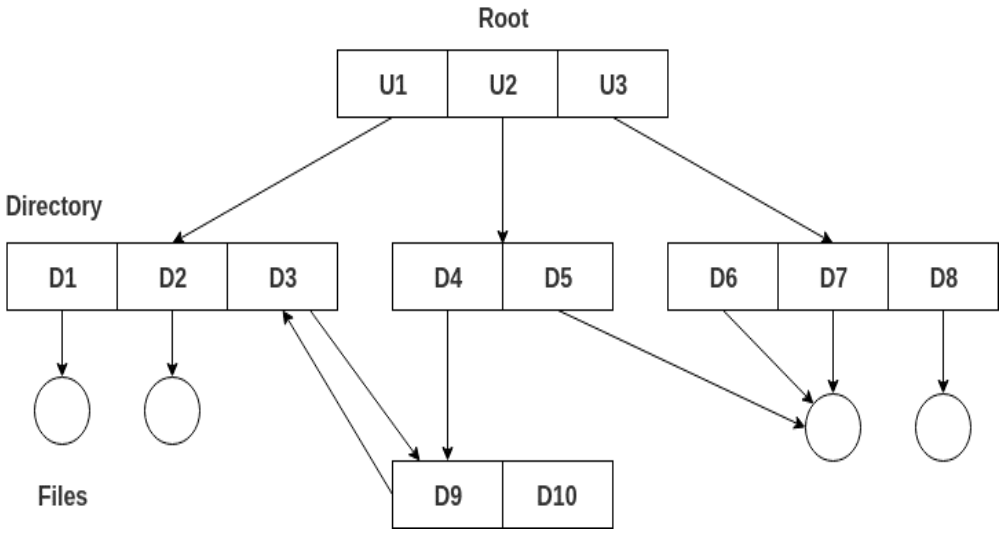
- We can share files.
- Searching is easy due to different-different paths.

Disadvantages:

- We share the files via linking, in case deleting it may create the problem,
- If the link is a soft link then after deleting the file we left with a dangling pointer.
- In the case of a hard link, to delete a file we have to delete all the references associated with it.

Generalgraphdirectorystructure–

In general graph directory structure, cycles are allowed within a directory structure where multiple directories can be derived from more than one parent directory. The main problem with this kind of directory structure is to calculate the total size or space that has been taken by the files and directories.



Advantages:

- It allows cycles.
- It is more flexible than other directories structure.

Disadvantages:

- It is more costly than others.
- It needs garbage collection.

8. What are the various file allocation methods? Explain with example [Nov 2018]

The allocation methods define how the files are stored in the disk blocks. There are three main disk space or file allocation methods.

- Contiguous Allocation
- Linked Allocation
- Indexed Allocation

The main idea behind these methods is to provide:

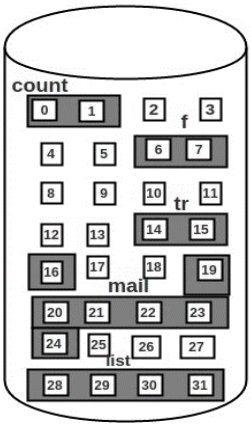
- Efficient disk space utilization.
- Fast access to the file blocks.

All the three methods have their own advantages and disadvantages as discussed below:

1. Contiguous Allocation

In this scheme, each file occupies a contiguous set of blocks on the disk. For example, if a file requires n blocks and is given a block b as the starting location, then the blocks assigned to the file will be: b, b+1, b+2,……b+n-1.

This means that given the starting block address and the length of the file (in terms of blocks required), we can determine the blocks occupied by the file. The directory entry for a file with contiguous allocation contains Address of starting block Length of the allocated portion. The file 'mail' in the following figure starts from the block 19 with length = 6 blocks. Therefore, it occupies 19, 20, 21, 22, 23, 24 blocks.



Directory

file	start	length
count	0	2
tr	14	3
mail	19	6
list	28	4
f	6	2

Advantages:

- Both the Sequential and Direct Accesses are supported by this. For direct access, the address of the kth block of the file which starts at block b can easily be obtained as (b+k).
- This is extremely fast since the number of seeks are minimal because of contiguous allocation of file blocks.

Disadvantages:

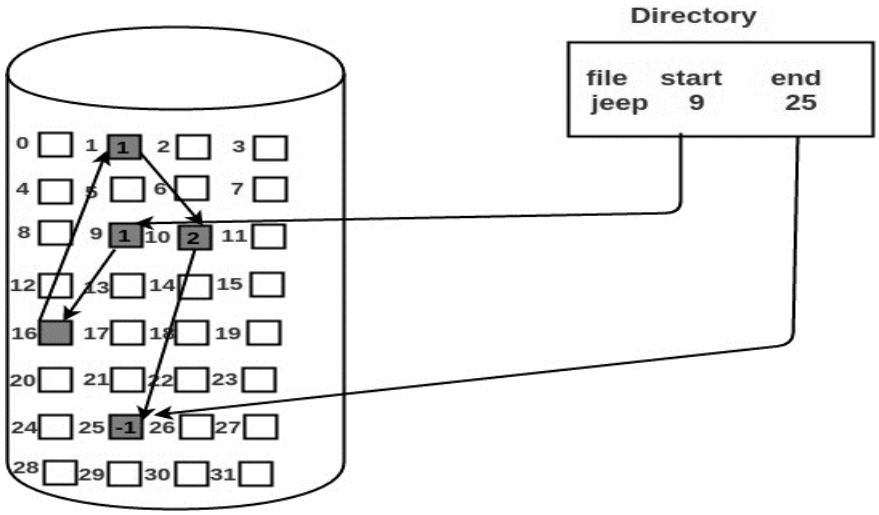
- This method suffers from both internal and external fragmentation. This makes it inefficient in terms of memory utilization.
- Increasing file size is difficult because it depends on the availability of contiguous memory at a particular instance.

2. Linked List Allocation

In this scheme, each file is a linked list of disk blocks which **need not be** contiguous. The disk blocks can be scattered anywhere on the disk. The directory entry contains a pointer to the starting and the ending file block. Each block contains a pointer to the next block occupied by the file.

The file 'jeep' in following image shows how the blocks are randomly distributed. The last block (25) contains -1 indicating a null pointer and does not point to any other

block.



Advantages:

- This is very flexible in terms of file size. File size can be increased easily since the system does not have to look for a contiguous chunk of memory.
- This method does not suffer from external fragmentation. This makes it relatively better in terms of memory utilization.

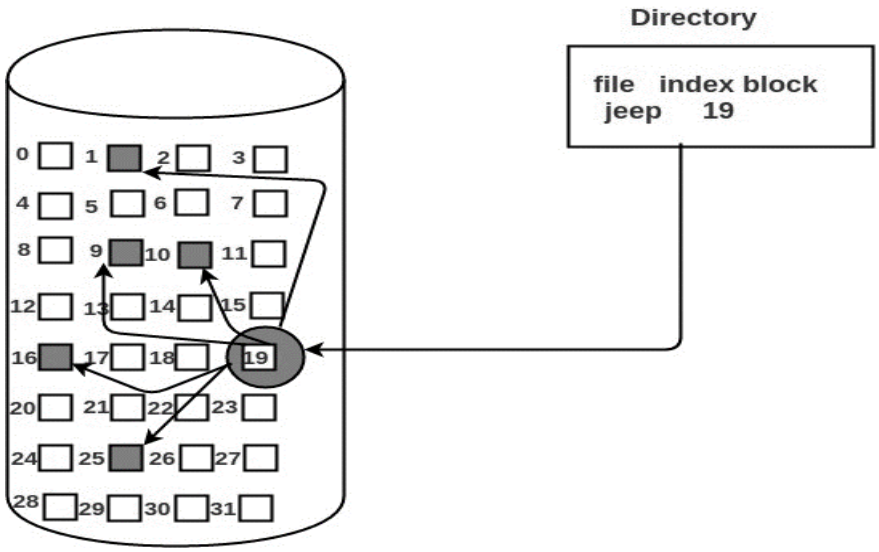
Disadvantages:

- Because the file blocks are distributed randomly on the disk, a large number of seeks are needed to access every block individually. This makes linked allocation slower.
- It does not support random or direct access. We can not directly access the blocks of a file. A block k of a file can be accessed by traversing k blocks sequentially (sequential access) from the starting block of the file via block pointers.
- Pointers required in the linked allocation incur some extra overhead.

3. Indexed Allocation

In this scheme, a special block known as the **Index block** contains the pointers to all the blocks occupied by a file. Each file has its own index block. The ith entry in the index block contains the disk address of the ith file block.

The directory entry contains the address of the index block as shown in the image:



Advantages:

- This supports direct access to the blocks occupied by the file and therefore provides fast access to the file blocks.
- It overcomes the problem of external fragmentation.

Disadvantages:

- The pointer overhead for indexed allocation is greater than linked allocation.
- For very small files, say files that expand only 2-3 blocks, the indexed allocation would keep one entire block (index block) for the pointers which is inefficient in terms of memory utilization. However, in linked allocation we lose the space of only 1 pointer per block.
- For files that are very large, single index block may not be able to hold all the pointers.

Following mechanisms can be used to resolve this:

Linked scheme:

This scheme links two or more index blocks together for holding the pointers. Every index block would then contain a pointer or the address to the next index block.

Multilevel index:

In this policy, a first level index block is used to point to the second level index blocks which inturn points to the disk blocks occupied by the file. This can be extended to 3 or more levels depending on the maximum file size.

Combined Scheme:

In this scheme, a special block called the Inode (information Node) contains all the information about the file such as the name, size, authority, etc and the remaining space of Inode is used to store the Disk Block addresses which contain the actual file as shown in the image below. The first few of these pointers in Inode point to the direct blocks i.e the pointers contain the addresses of the disk blocks that contain data of the file. The next few pointers point to indirect blocks.

Indirect blocks may be single indirect, double indirect or triple indirect. Single Indirect block is the disk block that does not contain the file data but the disk address of the blocks that contain the file data. Similarly, double indirect blocks do not contain the file data but the disk address of the blocks that contain the address of the blocks containing the file data.